

Crowded Carry in G10 FX and Crypto

Crash risk in FX and crypto carry trades

STUDENTS

Reza Zamani • **Paraj Goyal**

PROFESSOR

Amir Kermani

Currency Markets (MFE 230GB) • Final project • October 2026

- Our project asks one question: do carry trades crash when they are crowded?
- We test it in two markets with the same economics: G10 currencies and crypto.
- Strategy A is carry in seven G10 currencies against the US dollar, with a crowding filter built from CFTC data.
- Strategy C is funding carry in twelve crypto perpetual futures on Binance, including the coins that collapsed, Terra and FTX Token.

Carry pays because it crashes

Our question: do carry trades crash when they are crowded?

- We test it in two markets with the same economics: G10 currencies and crypto.
- Strategy A is carry in seven G10 currencies against the US dollar, with a crowding filter built from CFTC data.
- Strategy C is funding carry in twelve crypto perpetual futures on Binance, including the coins that collapsed, Terra and FTX Token.

Why carry crashes

- Carry borrows the low-yield asset and holds the high-yield one. UIP says this earns nothing, but in the data it earns a premium, with rare large crashes.
- **Carry goes "up the stairs and down the elevator": small steady gains most of the time, then large losses in a few weeks.**
- Crashes are worst when the trade is crowded and everyone unwinds at once (Brunnermeier, Nagel & Pedersen 2008).

Hypotheses, fixed before testing

- A: carry earns a premium, crowded months are followed by weaker carry, and it loses in volatility spikes.
- C: funding carry earns a premium with little BTC exposure, and its crashes are coin-specific.

A: G10 FX

C: Crypto

Carry signal	3-month rate differential	Perpetual funding rate
Crowding data	CFTC speculator positions	Funding = leveraged demand
Universe	7 currencies vs USD	12 coins (listed below)
Frequency	Monthly, 1999–2026	Weekly, 2020–2026

Currencies: Japanese yen (JPY), euro (EUR), British pound (GBP), Swiss franc (CHF), Canadian dollar (CAD), Australian dollar (AUD) and New Zealand dollar (NZD), each against the US dollar.

Coins: Bitcoin (BTC), Ethereum (ETH), BNB, XRP, Cardano (ADA), Dogecoin (DOGE), Solana (SOL), Litecoin (LTC), Chainlink (LINK), Avalanche (AVAX), Terra (LUNA) and FTX Token (FTT).

1

- Carry means borrowing in a low-yield asset and holding a high-yield one.
- Uncovered interest parity says this should earn nothing, but in the data it earns a premium.
- The catch is crash risk: carry goes up the stairs and down the elevator.
- Brunnermeier, Nagel and Pedersen show crashes are worst when speculators are crowded into the trade.
- In FX the yield is the interest-rate differential, and we measure crowding with CFTC speculator positions.
- Our FX universe is the yen, euro, British pound, Swiss franc, Canadian dollar, Australian dollar and New Zealand dollar, all against the US dollar.
- In crypto the yield is the perpetual funding rate, which also measures leveraged demand.
- Our coins are Bitcoin, Ethereum, BNB, XRP, Cardano, Dogecoin, Solana, Litecoin, Chainlink, Avalanche, Terra and FTX Token.
- We wrote the hypotheses and all parameters down before running any backtest.
- Uncovered interest parity (UIP) says a high-rate currency should fall by exactly the rate gap, so carry would earn zero; in practice high-rate currencies tend not to fall that much, which is the forward premium puzzle.
- 'Up the stairs and down the elevator' means carry earns small steady gains most of the time and then loses a lot in a few weeks, so its returns have negative skew.
- Crowding matters because when many speculators hold the same trade, a shock forces them all to unwind together, which makes the crash larger.
- Hypothesis A says three things: carry earns a premium, crowded months are followed by weaker carry, and carry loses when volatility spikes.
- Hypothesis C says crypto funding carry earns a premium with little exposure to Bitcoin, and that its crashes come from single coins.

How the two strategies trade

Each step uses only data available at the time of the trade

Strategy A: G10 FX carry, monthly

- 1 **Month-end: rank**
Compute each currency's 3-month rate minus the US rate.
- 2 **Trade**
Buy the 2 highest-rate currencies and sell the 2 lowest, 50% each, so there is no net dollar bet.
- 3 **Crowding check**
Z-score CFTC speculator positions over 36 months; if crowding is above 1, halve all positions.
- 4 **Hold one month**
Earn the currency move plus the interest-rate gap.
- 5 **Rebalance**
At the next month-end, rank again and swap, paying 3 bp per trade.

Strategy C: crypto funding carry, weekly

- 1 **Sunday close: rank**
Average each coin's daily funding rate over the past 7 days, using data up to the day before.
- 2 **Trade**
Short the third of coins with the highest funding and buy the lowest third (usually 3 each), so the book is dollar-neutral.
- 3 **Hold 7 days**
Earn the price move plus funding; a short receives funding when it is positive.
- 4 **Rebalance**
Next Sunday, rank again and swap, paying 5 bp per trade.

2

- Here is how each strategy trades, step by step.
- Strategy A trades once a month: at each month-end we rank the seven currencies by their interest rate against the US dollar, buy the top two and sell the bottom two.
- Before trading we check crowding from CFTC data, which we use only after its Friday release; if crowding is above one, we halve the positions.
- We hold for one month and then rank again.
- Strategy C trades every Sunday: we rank the coins that are trading by their average funding over the past week, short the third with the highest funding and buy the third with the lowest, which is usually three coins on each side.
- We hold for seven days, collecting funding on the shorts, and then rank again.

Data and methodology

Public data; a research design chosen to keep the tests honest

Data	Source	Coverage
FX spot, 7 currencies	FRED (Fed H.10)	1999–2026
3-month rates	OECD via FRED	1999–2026
Speculator positions	CFTC Commitments of Traders	1986–2026
Crypto price, volume, funding	Binance public archive	2020–2026
Carry factor (validation)	Lustig–Roussanov–Verdelhan	1983–2021

Data fix: before Aug 2000 CFTC lists the same currency futures under 'International Monetary Market'. Matching both names extended positioning back to 1986.

- All parameters were fixed before the first backtest.
- The out-of-sample period starts in 2011 for A and in 2024 for C.
- Costs are 3 bp per unit traded in FX and 5 bp in crypto.
- There is no look-ahead: weights set at t earn the return of $t+1$, and CFTC data are used only after their Friday release.
- There is no survivorship bias, because Terra (LUNA) and FTX Token (FTT) stay in the sample.
- Robustness grids are reported, but they were never used for tuning.

3

- All data are public: FRED for FX and rates, CFTC for positions, Binance for crypto.
- One data fix worth mentioning: before 2000 the CFTC lists currency futures under a different exchange name; matching both names gave us positioning back to 1986.
- Every parameter was fixed in advance, and our out-of-sample period starts in 2011 for FX and in 2024 for crypto.
- Returns are net of costs, there is no look-ahead, and we keep the two coins that died, Terra (LUNA) and FTX Token (FTT).
- Out-of-sample means the later years that we did not look at while designing the strategy; if a rule still works there, it is less likely to be a result of data mining.
- Look-ahead bias means using information that was not public at the time of the trade; we avoid it because weights set at t earn the return of $t+1$, and CFTC positions measured on Tuesday are used only after their Friday release.
- Survivorship bias means testing only on assets that still exist today; we avoid it by keeping Terra and FTX Token, which together cost the crypto strategy about 100 percentage points of P&L.
- A basis point (bp) is 0.01%, so a cost of 3 bp means we lose 0.03% of every dollar we trade.
- The Lustig–Roussanov–Verdelhan (LRV) factor is the published academic carry factor, built by buying high-rate and selling low-rate currency portfolios; we use it only to check our own carry series.

Strategy A: crowding-filtered G10 carry

Yen, euro, pound, Swiss franc, Canadian, Australian and New Zealand dollar vs USD, monthly

1

Carry portfolio

At each month-end we buy the 2 highest-rate currencies and sell the 2 lowest, with equal weights and no net dollar bet.

2

Crowding signal

Each currency's CFTC net speculative position over open interest is z-scored over 36 months.

3

Portfolio crowding

It is the average z of the long leg minus the average z of the short leg.

4

Filter

If crowding is above 1, all positions are cut to half for the next month.

33

crowded months out of 328, 1999–2026

Who we usually hold

- Long: New Zealand dollar (94% of months) and Australian dollar (75%).
- Short: Swiss franc (99%), Japanese yen (52%) and euro (40%).

- Each month we rank the seven currencies by their three-month interest rate.
- We buy the two highest and sell the two lowest, so the portfolio has no net dollar bet.
- In practice we are almost always long the New Zealand and Australian dollars and short the Swiss franc, often with the yen or the euro.
- For crowding we take speculators' net futures position and z-score it over 36 months.
- Crowding is high when speculators are unusually long what we hold and short what we sell.
- When this score is above one standard deviation, we halve the position, which happened in 33 months.

Crowding filter helps out of sample, not in 2008



0.27 → 0.32

Out-of-sample Sharpe, net: carry vs crowd-filtered (2011–2026)

0.30% vs 0.02%

Next-month carry after non-crowded vs crowded months

0.82

Correlation with the academic LRV carry factor (1999–2021)

Net, 1999–2026	Net return a year	Volatility a year	Max drawdown
Carry	3.3%	8.9%	-37%
Crowd-filtered	3.2%	8.5%	-37%

5

- Plain carry earns about 3.3% a year after costs with 8.9% volatility, a net Sharpe of 0.37.
- Our carry has a 0.82 correlation with the published Lustig–Roussanov–Verdelhan factor, which validates the code and data.
- After crowded months, next-month carry averages only 0.02% versus 0.30% otherwise, with more negative skew.
- Out of sample the filter lifts the Sharpe from 0.27 to 0.32, and every variant in our robustness grid also beats 0.27.
- But it did not avoid 2008: the crash started from a non-crowded reading, so the max drawdown is -37% either way.
- One caveat is that there are only 33 crowded months, so the effect is not precisely estimated.

Strategy A loses when volatility spikes

Monthly net carry on risk factors, Newey-West t-stats, 1999–2021

-1.55

Beta to equity-volatility changes (t = -3.7)

0.35

Beta to the dollar factor (t = 4.1)

0.25%

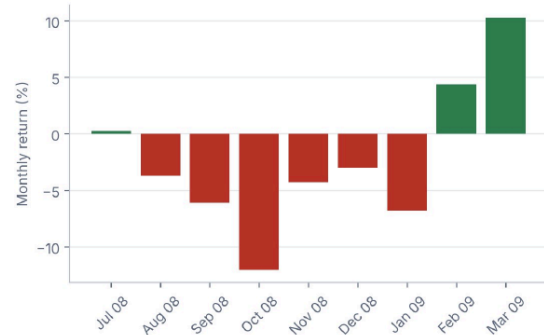
Monthly alpha (t = 1.7): not significant

0.18

R² of the two-factor model

Worst months: Oct 2008, Aug 2007, May 2000, Jan 2009, and the Jan 2015 Swiss franc de-peg.

The 2008 carry crash: -20.5% (Jul 2008 – Mar 2009)



- To see when it loses, we regress monthly carry on the dollar factor and on equity-volatility shocks.
- The volatility beta is -1.55 with a t-stat of -3.7: carry loses when volatility jumps.
- Once we control for these factors the alpha is not significant, so carry is pay for risk, not a free lunch.
- From July 2008 to March 2009 the strategy lost 20.5% as the funding currencies, the Japanese yen and the Swiss franc, rallied against the Australian and New Zealand dollars.

Strategy C: crypto funding carry

Binance perpetuals on BTC, ETH, BNB, XRP, ADA, DOGE, SOL, LTC, LINK, AVAX, LUNA (Terra), FTT; weekly, 2020–2026



- Funding is paid every 8 hours, and positive funding means longs pay shorts.
- Each Sunday we short the third of coins with the highest 7-day funding and buy the lowest third.
- The book is dollar-neutral and pays 5 bp per trade.

0.14

Net Sharpe, full

0.81

Net Sharpe, 2024–26

-27%

Worst week: LUNA (Terra) collapse, May 2022

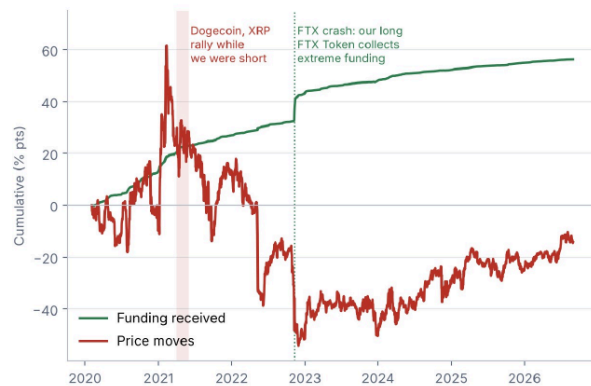
Net of costs	Net return a year	Volatility a year	Max drawdown
Full sample 2020–26	4.1%	30%	-67%
Since 2024	13.9%	17%	-13%

7

- Crypto perpetual futures have their own interest rate, the funding rate, paid every 8 hours between longs and shorts.
- High funding means many leveraged longs, so it is both a carry signal and a crowding signal.
- Each Sunday we short the third of coins with the highest funding and buy the third with the lowest.
- The twelve coins are Bitcoin, Ethereum, BNB, XRP, Cardano, Dogecoin, Solana, Litecoin, Chainlink, Avalanche, Terra and FTX Token.
- We keep Terra (LUNA) and FTX Token (FTT), which both collapsed in 2022, to avoid survivorship bias.
- Net Sharpe is 0.14 over the full sample: -0.07 in 2020 to 2023 and 0.81 since 2024.
- After costs it earns 4.1% a year with 30% volatility, and its maximum drawdown is -67%, much deeper than FX carry.
- The worst week was the Terra (LUNA) collapse in May 2022, -27.3%: traders were shorting Terra, its funding turned negative, and so our rule held it long.
- A perpetual future has no expiry date, so the exchange uses the funding payment to keep its price close to the spot price.
- For example, if funding is +0.01% every 8 hours, longs pay shorts about 0.03% a day, which is roughly 11% a year.
- So shorting a coin with high funding earns that payment, just as selling a low-rate currency and buying a high-rate one earns the interest gap in FX.
- The book is dollar-neutral, which means the long and short sides have the same size, so a move of the whole crypto market should roughly cancel out.
- The out-of-sample Sharpe since 2024 is strong, but it comes from less than three years of data, so we treat it with caution.

Funding is earned, but price moves dominate

Where the return comes from, and what drives the losses



Total P&L by coin, % pts: Solana +102, BNB +51, Avalanche +36; Terra -87, XRP -58, Dogecoin -43, FTX Token -13.

Robustness: all 6 lookback and rebalancing settings have a positive out-of-sample Sharpe (0.41 to 1.29); our pre-set choice is in the bottom half.

% per year	Funding	Price	Net
Full	+8.5	-2.2	+4.1
In-sample	+12.3	-12.5	-2.5
Out-of-sample	+3.1	+13.0	+13.9

Takeaways

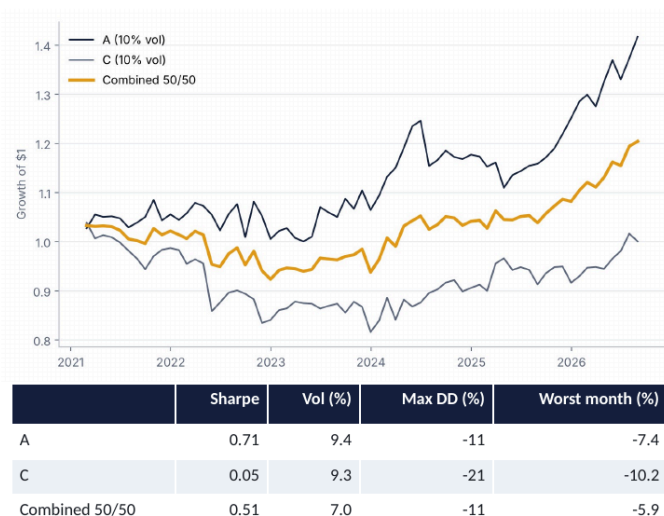
- Funding income is steady at about 8.5% a year, but price moves decide the result.
- The strategy has no market exposure, since its beta to Bitcoin is -0.02.
- Its losses come from single coins: Terra lost 87 points, XRP 58 and Dogecoin 43.
- Deeply negative funding signals crowded shorts, not a cheap coin.

8

- We split the return into funding and price. Funding adds about 8.5% a year, steadily.
- But price moves dominate: in 2020 to 2023 the coins we shorted, like Dogecoin and XRP, kept rallying and wiped out the funding.
- Since 2024 most of the gain came from price, so we read the strong out-of-sample number with caution.
- The beta to Bitcoin is about zero, so the strategy has no market exposure.
- Its losses come from single coins: Terra alone cost 87 percentage points, XRP 58 and Dogecoin 43, while Solana and BNB were the biggest winners.
- Our takeaway is that crypto carry crashes in the mirror image of FX: it ends up long whatever traders short hardest.
- As a robustness check we tried 6 lookback and rebalancing settings; all have a positive out-of-sample Sharpe, and the setting we fixed in advance is in the bottom half, so we did not pick the best one after the fact.
- The table adds up: over the full sample, funding of +8.5%, price of -2.2% and costs of -2.2% give a net return of +4.1% a year.
- In 2020 to 2023 funding paid 12.3% a year, but price moves took away 12.5%, so the strategy lost money.
- Since 2024 funding paid only 3.1%, and price moves added 13.0%, which is luck rather than carry.

Combining FX and crypto carry

Each scaled to 10% vol with lagged volatility, 50/50, 2021–2026



0.11

Correlation between A and C, monthly, 2021–2026

Takeaways

- FX carry and crypto carry are almost uncorrelated, so they crash at different times.
- The 50/50 mix has 7.0% volatility, compared with about 9% for each strategy alone.
- Its worst month is -5.9%, smaller than for FX (-7.4%) or crypto (-10.2%).
- The mix does not beat FX alone on Sharpe (0.51 vs 0.71), because crypto carry was weak in this window.

9

- Finally, we scale both strategies to 10% volatility and combine them 50/50.
- Their correlation is only 0.11: FX and crypto carry crash at different times.
- The mix has 7.0% volatility and a Sharpe of 0.51, and its worst month, -5.9%, is smaller than either part's.
- It does not beat FX alone on Sharpe, because crypto carry was weak in this window.
- Our takeaway is that the benefit of combining them is lower risk, not a higher return.
- We scale each strategy to 10% volatility so that neither one dominates the mix just because it is more volatile; crypto carry is about three times as volatile as FX carry before scaling.
- The scaling uses past volatility only, so there is no look-ahead in the combined portfolio.
- A correlation of 0.11 is close to zero, which means a bad month in one strategy says very little about the other.
- The window starts in 2021 because that is when both strategies have enough history for the volatility estimate.
- With only 67 months of overlap, these combined numbers are indicative rather than precise.

Conclusions and limitations

Our question: do carry trades crash when they are crowded?

Short answer: partly yes. Crowding makes carry weaker, but it doesn't predict the biggest crashes.

- FX: carry crashes are compensation for risk, and crowding is a modest warning sign, not a reliable crash predictor.
- Crypto: carry also earns a premium and also crashes, but the crashes come from single coins, mostly being long a coin that traders are crowding to short, so crowding tells you where the danger is rather than when it will hit.

Hypotheses: verdict and evidence

Hypothesis	Verdict	Evidence
A1: G10 carry earns a premium	Supported	Net Sharpe 0.37; 0.82 correlation with LRV
A2: crowding predicts weaker carry	Leaning yes	0.02% vs 0.30% next month; OOS Sharpe 0.27 → 0.32; missed 2008
A3: carry loses in volatility spikes	Supported	Vol beta -1.55 ($t = -3.7$); alpha not significant
C1: funding carry earns a premium	Mixed	Funding +8.5% a year, but net Sharpe 0.14; price moves dominate
C2: no BTC exposure; coin-specific crashes	Supported	BTC beta = 0; Terra -87 pts; worst week -27%
FX and crypto carry diversify	Supported	Correlation 0.11; vol 7.0% vs about 9%

What is new

- We use CFTC speculator positioning as a crowding filter on G10 carry.
- We treat crypto funding as the crypto interest-rate gap and as a crowding signal, and test it with the coins that died.

Limitations

- The crypto history is short (2020–2026) and covers only 12 coins.
- There are only 33 crowded FX months, and the 2008 crash was not flagged.
- Costs are simple estimates, and we ignore funding-rate caps, borrow limits and exchange risk.

10

- Our short answer is partly yes: crowding makes carry weaker, but it doesn't predict the biggest crashes.
- In FX, carry crashes are compensation for risk, and crowding is a modest warning sign, not a reliable crash predictor.
- In crypto, carry also earns a premium and also crashes, but the crashes come from single coins, mostly being long a coin that traders are crowding to short, so crowding tells you where the danger is rather than when it will hit.
- The table sums up our hypotheses: four are supported, crowding in FX leans yes, and crypto funding carry is mixed.
- In both markets, carry looks like compensation for crash risk.
- In FX, crowding measured from CFTC positions is a cheap and modestly useful warning sign.
- In crypto, sorting on funding alone exposes you to single-coin collapses like Terra and FTX Token.
- Because the two crash at different times, they diversify each other.
- What is new in our work is using CFTC positioning as a crowding filter on carry, and treating crypto funding as both the crypto interest-rate gap and a crowding signal, tested with the coins that died.
- The main limitations are a short crypto sample, few crowded FX months, and simple cost estimates.